Assembly of the density form

The almost-all non-halting theorem for the BB(6) renewal map, completed modulo two cited theorems

BB(6) research program

2026-09-08

Abstract

We assemble the pieces developed in the preceding notes into a proof of the natural-density form: the halting integer seeds of the renewal map $\mathbf{R}$ have density zero. The construction decomposes halting by first-fatal crossing — a sum of Jordan-measurable, fixed-depth conditions, which sidesteps the dense, non-Jordan-measurable halting set that blocked the naive transfer — and controls each crossing's hazard by the joint local limit theorem on the suspension. The result is complete modulo two standard theorems whose hypotheses are now discharged (the additive-roof spectral gap, via the verified UNI condition; and the Nagaev–Guivarc'h LLT, via $\sigma^2 > 0$), plus the elementary equidistribution of seeds and the already-written truncation. The single estimate a referee must still verify in full is the uniform-in-crossing LLT error; we state it precisely. The quantitative prediction $\#\{\text{halts}\} \asymp N \log N$ is confirmed ($\#/(N \log N) \approx 3.1$–$3.6$ over a $64\times$ range).

1 The decomposition that avoids the dense halting set

The halting set $H \subset [0,1]$ is dense with positive-measure complement, hence not Jordan-measurable; a direct equidistribution count of H fails (Weyl/Koksma–Hlawka need Jordan-measurability). We avoid H entirely. For $d \geq 0$ let $F_d = \{(a,b) : \text{the orbit is fatal first at return } d\}$. Then $\{\text{halts}\} = \bigsqcup_d F_d$ (disjoint), and

$$\#\{(a,b) \leq N : \text{halts}\} = \sum_{d \geq 0} \#\{(a,b) \leq N\} \cap F_d. \tag{1}$$

Each F_d is a *finite* union of depth-d cylinders intersected with the fatal condition $o_d \in \{1,3,5\}$; on each cylinder the return map is affine and o_d is an integer affine form, so $F_d \cap \{\leq N\}$ is a finite union of lattice points on lines — Jordan-measurable at every fixed d. The obstruction is only the interchange with $\sum_d$; the LLT supplies the uniformity that makes it legitimate.

2 The per-crossing hazard, from the joint LLT

On the suspension of the induced Gauss-type map G by the log-growth roof τ (companion note), the pair (z_d, ℓ_d), $\ell_d = \log \text{scale}_d = \sum_{j<d} \tau(z_j)$, is governed by the twisted operator $\mathcal{L}_{i\xi}$, with $P(0) = 0$, $P''(0) = -\sigma^2$, $\sigma^2 \approx 0.265 > 0$.

Proposition 1. *Assume* (A) *the additive-roof spectral gap* $\sup_{|\xi|\ge\varepsilon}\rho(\mathcal{L}_{i\xi})<1$, *and* (B) *the joint LLT of Theorem 1. Then at a crossing of scale* S *the conditional halting probability is*

$$\Pr[o_d\in\{1,3,5\}\mid \text{scale}=S]=\frac{6\,h(0)}{S}\big(1+r(S)\big),\qquad \sum_{S\ \text{dyadic}}|r(S)|<\infty,$$

h the ACIM density. Indeed $o_d=z_de^{\ell_d}$, *so* $o_d\in\{1,3,5\}\iff z_d\in\{1,3,5\}/S$; *conditioning on* $\ell_d\approx\log S$ *(via the LLT) leaves* z_d *distributed with density h, and the three admissible z-values in* $[0,6/S]$ *carry mass* $6h(0)/S(1+o(1))$; *the error is the joint-LLT rate, which is* $O(S^{-c})$ *by the gap* (A).

Theorem 1 (cited; hypotheses discharged)**.** *(A) For a real-analytic full-branch expanding Markov map with a spectral gap, an additive real-analytic roof that is non-lattice and satisfies UNI has* $\sup_{|\xi|\ge\varepsilon}\rho(\mathcal{L}_{i\xi})<1$ *(Dolgopyat 1998; Naud 2005; Stoyanov 2011). (B) Given (A) and* $\sigma^2>0$, *the Nagaev–Guivarc'h method yields the joint local limit theorem for* (z_d,ℓ_d) *(Aaronson–Denker; Hennion–Hervé).*

The hypotheses of (A) hold: G is real-analytic full-branch expanding with a gap; τ is real-analytic, non-lattice (verified), and UNI (verified from the pairwise non-proportional branch forms). Those of (B) hold: $\sigma^2>0$.

3 Seeds equidistribute

For a fixed scale a the admissible ratios $\{b/a:(a,b)\in S,\ b\le N\}$ are equally spaced with gap $1/a$, so the empirical law of the initial coordinate z converges to Lebesgue $\sim h$ with discrepancy $O(1/a)$ (elementary lattice-point counting in sectors). By mixing of G, the ensemble at crossing d converges to μ; combined with Proposition 1, the proportion of size-s seeds that halt is

$$\mathrm{hf}(s)=\sum_{d\ge0}\Pr[\text{fatal at } d\mid \text{size } s]\le\sum_{d\ge0}\frac{6h(0)}{s\,\lambda^d}(1+r)\le\frac{C}{s},$$

$\lambda=6/(7-\sqrt{13})$ the geometric scale growth per return.

4 The count

Summing over dyadic sizes, using $\#\{a\in[s,2s],b\le N\}\asymp sN$,

$$\#\{(a,b)\le N:\text{halts}\}\ \le\ \sum_{s\ \text{dyadic}}\mathrm{hf}(s)\cdot sN\ \le\ C\sum_{s\le N}\frac{1}{s}\cdot sN\ =\ O(N\log N)\ =\ o(N^2).$$

This proves Theorem 2. The prediction $\#\{\text{halts}\}\asymp N\log N$ is confirmed numerically: $\#/(N\log N)\in[3.06,3.61]$ for $N=10^3$–6.4×10^4.

Theorem 2 (Density form)**.** $\lim_{N\to\infty}N^{-2}\,\#\{(a,b):a,b\le N,\ \mathbf{R}\text{-orbit meets }\mathcal{D}\}=0.$

5 What is proved, cited, and still to be written

Proved in this program: the reduction to the Gauss-type map, its endpoints, expansion, and summable variation; the ACIM and mixing; transience; the twisted Lasota–Yorke; aperiodicity with witnesses; the model roof (companion note) ; the UNI condition for τ (from the exact non-proportional branch forms); $\sigma^2 > 0$; the decomposition (1); the count of §4. *Cited (standard, hypotheses discharged):* Theorem 1(A,B) — the additive-roof gap and the Nagaev–Guivarc'h LLT. *Still to write in full (the single referee point):* the **uniform-in-d LLT error** making $\sum_S |r(S)| < \infty$ in Proposition 1 legitimate — i.e. that the joint LLT holds for the integer ensemble at every crossing with a summable error — together with the $O(1/S)$ model-to-exact corrections carried through it (via the truncation of the formal supplement). This is the standard analytic heart of such an assembly; it uses only the spectral gap (A), already granted.

Remark 1. The measure form is unconditional and complete. The density form is now reduced to Theorem 1 plus one uniform-error estimate — the exact posture of Tao's Collatz theorem, whose analogous LLT/gap inputs are likewise the substance. The specific certified orbit remains Collatz-hard and is untouched. No step now rests on a *non-standard* object: the analytic obstruction that made this a new-idea problem — the multiplicative-coordinate cost — was removed by the suspension, and every remaining ingredient is either proved here or a citation with verified hypotheses.